REVISION NOTES

Week 4

Sun 8 – Fri 13 November 2026

14	10 h 30 min	14 / 14
LECTURES	VIDEO	SUMMARISED

COURSES THIS WEEK

MATH 120	Single-Variable Calculus	2 lectures · 1 h 36 min
AI 200	Neural Networks: Zero to Hero	3 lectures · 4 h 20 min
SPCH 100	Storytelling	5 lectures · 1 h 18 min
MATH 130	Probability	2 lectures · 1 h 42 min
MATH 110	Linear Algebra	2 lectures · 1 h 34 min
Total		**14 lectures · 10 h 30 min**

Study days Sun 8 Nov to Fri 13 Nov. Hours are video time; a lecture the plan spreads over several days counts only the days that fall in this week.

WEEK 4 · SUN 8 – FRI 13 NOVEMBER 2026

Contents

LECTURES, IN PLAN ORDER

1	MATH 120	Derivatives	Sun 8
2	AI 200	Let's build GPT — from scratch, in code	Sun 8 – Mon 9
3	SPCH 100	Promise, progress, payoff — why a middle drags and an ending lands (start at 19:28)	Sun 8 – Tue 10
4	MATH 130	Gambler's Ruin and Random Variables	Mon 9
5	MATH 110	Transposes, Permutations, Spaces R^n	Tue 10
6	AI 200	Let's build the GPT Tokenizer	Tue 10 – Thu 12
7	MATH 120	Chain Rule	Wed 11
8	SPCH 100	Homework for Life, sharpened — and the pickle story	Wed 11
9	SPCH 100	Seeing with storyteller eyes — notice the flag	Wed 11
10	SPCH 100	A photo folder called Stories	Wed 11
11	MATH 130	Random Variables and Their Distributions	Thu 12
12	SPCH 100	First, last, best, worst — mining the past	Thu 12
13	MATH 110	Column Space and Nullspace	Fri 13
14	AI 200	Let's reproduce GPT-2 (124M)	Fri 13

AT THE BACK

What this week established

Answers to the self-checks

Every lecture this week has a summary.

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 1 OF 14

Derivatives

Sun 8 Nov · 50 min video · Watch on YouTube

THE TAKEAWAY

Both trig derivatives come from two limits at zero, and those two limits come from one geometric fact: short pieces of curve are nearly straight.

THE ARGUMENT · 6 STEPS

1. **Two kinds of formula, and you need both**
   SPECIFIC ones give a named function's derivative — x^n, $1/x$, $\sin x$. GENERAL ones combine what you have — sums, constant multiples, and soon products and quotients. A handful of each generates everything.

2. **Expand with the sum formula, then group to keep $0/0$ together**
   $\sin(x+\Delta x) = \sin x \cos \Delta x + \cos x \sin \Delta x$. Subtracting $\sin x$ and grouping gives $\sin x \cdot \frac{\cos \Delta x - 1}{\Delta x} + \cos x \cdot \frac{\sin \Delta x}{\Delta x}$. The grouping is the whole trick: a term that is not over a zero becomes meaningless.

3. **Two limits do all the work**
   (A) $\frac{\cos \Delta x - 1}{\Delta x} \to 0$ and (B) $\frac{\sin \Delta x}{\Delta x} \to 1$. So $\frac{d}{dx} \sin x = \cos x$, and the same expansion for cosine gives $\frac{d}{dx} \cos x = -\sin x$.

4. **Those two limits ARE the derivatives at zero**
   (B) is $\frac{d}{dx} \sin x$ at $x = 0$; (A) is $\frac{d}{dx} \cos x$ at $x = 0$. So knowing the rate of change at one single place delivers it at every place — that is the structure of the proof.

5. **Why (B) is 1: the bow and the bowstring**
   On the unit circle, $2 \sin \theta$ is the chord and 2θ is the arc. As θ closes, the curve and the straight segment merge in length, so their ratio goes to 1. Short curves are nearly straight.

6. **Why (A) is 0: the gap closes faster**
   $1 - \cos \theta$ is the tiny gap between the bowstring and the bow, and θ is the arc. Both go to zero, but the gap goes far faster — so the ratio goes to 0, not to 1.

WORKED EXAMPLE

A second proof, for all θ at once: move a point round the unit circle by $\Delta\theta$. The chord has length about $\Delta\theta$ and is nearly perpendicular to the radius, so the angle in the little triangle is again θ. Hence $\Delta y \approx \Delta\theta \cos\theta$.

WATCH FOR

Using degrees. Every one of these arguments compares an ARC LENGTH with a vertical distance, so the angle must be measured as length along the unit circle. In degrees the formulas are simply wrong.

CHECK YOURSELF · ANSWERS AT THE BACK

1.1 $\frac{d}{dx}\sin x = \cos x$ holds only when x is in radians because:

A Radians are more modern
B The proof compares the arc length θ with a chord, so the angle must be measured as length on the unit circle
C Degrees make sine negative
D Cosine is undefined in degrees

1.2 The derivative of $\sin x$ at $x = \pi/3$ is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 2 OF 14

Let's build GPT — from scratch, in code

Sun 8 Nov – Mon 9 Nov · day 2–3 of 3 · 116 min video · Watch on YouTube · began in week 3

THE TAKEAWAY

A GPT is a bigram model that learned to look back: tokens gather information from earlier positions through masked, scaled self-attention, compute on it with a feed-forward net, and stack those blocks with residual connections and layer norms — each addition visibly lowering the validation loss on Tiny Shakespeare.

THE ARGUMENT · 8 STEPS

1. **Characters as tokens, chunks as batches**
Tiny Shakespeare is encoded one character per integer, split 90 / 10 into train and val. get_batch draws random chunks: x is block_size tokens and y the same chunk shifted by one, so one chunk holds block_size examples, with contexts of length 1 up to block_size.

2. **The baseline: a bigram model**
Each token reads the logits for the next token straight out of an embedding table; generate() samples one token at a time from the last position's softmax and appends it. Trained with AdamW, it produces Shakespeare-shaped gibberish.

3. **Embeddings with positions**
Tokens get n_embd-dimensional embeddings and a linear lm_head to the vocabulary; a position embedding table adds where each token sits, since attention by itself has no notion of order.

4. **One head of self-attention**
Each position emits a query and a key; their dot products, scaled by $1/\sqrt{\text{head size}}$, are the affinities (the gpt.py published with the video scaled by the embedding size C instead; a fix after the video changed it). A lower-triangular mask sets the future to −inf, softmax turns each row into weights, and the output is a weighted sum of the values. Val loss: 2.5.

5 **Many heads, then computation**

Four heads of size 8, concatenated back to 32: several independent communication channels (val 2.28). A per-position feed-forward net — linear, ReLU — then lets each token compute on what it gathered (2.24). Attention is communication; the feed-forward net is computation.

6 **Residuals and layer norm make it deep**

Blocks are stacked with x = x + f(x), a projection back into the residual stream, and a 4x wider feed-forward layer; layer norm is applied to x before each sub-layer and once more before lm_head. Val loss 2.06.

7 **Scaling up**

batch 64, block_size 256, n_embd 384, 6 heads, 6 layers, learning rate 3e-4 and dropout 0.2 on the attention weights, the projections and the feed-forward output — and the samples in more.txt read like a play.

8 **Generation has a window**

generate() crops the running sequence to its last block_size tokens before every forward pass: the position table has only block_size rows, and the mask is that wide.

WORKED EXAMPLE

Track the shapes through one head: x is (B, T, C); q and k are (B, T, hs); q @ k.transpose(-2, -1) is (B, T, T) — one row of affinities per position; after the mask and softmax, @ v gives (B, T, hs). Then multi-head concatenates n_head of those back to C, which is why head_size = n_embd // n_head.

WATCH FOR

Forgetting the mask, or masking after the softmax. Without the −inf entries in the upper triangle, every position can attend to the tokens it is supposed to predict, the training loss collapses, and generation — which has no future to look at — falls apart.

CHECK YOURSELF · ANSWERS AT THE BACK

2.1 get_batch takes x = data[i : i+block_size] and y = data[i+1 : i+block_size+1]. With block_size = 8, how many (context, target) training examples does one such chunk pack?

Your answer: _____

2.2 In a self-attention head, why is wei filled with −inf wherever the lower-triangular tril matrix is 0, before the softmax?

A To stop the attention weights from growing too large during training
B To make the attention weights symmetric between any two positions
C So each position gathers only from itself and earlier positions, never later
D So that the softmax output will sum to the head size, rather than to exactly one

2.3 The lecture scales the attention scores by 1/√(head size), so that they keep roughly unit variance before the softmax. With n_embd = 384 and n_head = 6, by what factor are the scores q · k multiplied? Three decimals.

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 3 OF 14

Promise, progress, payoff — why a middle drags and an ending lands (start at 19:28)

Sun 8 Nov – Tue 10 Nov · day 1–3 of 3 · 56 min video · Watch on YouTube

THE TAKEAWAY

Brandon Sanderson reduces plot to promise, progress and payoff. The opening promises what kind of story it is; the middle must keep showing movement toward that promise; the ending fulfils it in a way that feels surprising yet inevitable. When an audience is bored, the progress is usually on a different axis from the promise.

THE ARGUMENT · 6 STEPS

1. **Big P and little p**
   The big plot is what you can say in a sentence; the little plot is the small problem-and-solution beats that keep people engaged moment to moment.

2. **The tone promise**
   If it will be funny, there are jokes early; if it will be dark, the opening is dark. Names, descriptions and the first pages all make the promise.

3. **The story in miniature**
   Raiders of the Lost Ark opens with a mini adventure that is the whole film in small: a clever hero, hard trials, and he loses anyway.

4. **You control progress**
   You could end any story in one sentence or spend fifty pages on one second. The job is a satisfying sense of progress, signposted — 'we're getting closer, one more piece'.

5. **The side-quest problem**
   Promise point B, then send the characters to point C, and readers stop investing. His fix in Oathbringer was a new promise: if one character does not reach C, someone he loves will die.

6. **Surprising yet inevitable**
   Obstacles, escalation and red herrings make people doubt the promise. The toy-car-and-toy-plane twist makes them want something else, then gives it to them.

WORKED EXAMPLE

A story about nearly missing a job interview. Promise: 'I almost didn't make it to the interview that got me this job' — tense, a little comic, and the question is whether he makes it. Progress: the missed bus, the wrong building, the lift out of order, each told against the clock ('eleven minutes left'). Doubt: at the top of the stairs the receptionist says the panel has gone home. Payoff: one of them is stuck in that broken lift. A detour into what he had for breakfast would feel like a side quest — cut it, or tie it to the clock.

WATCH FOR

Start at about 19:28; the first nineteen minutes are course logistics and questions about outlining (the big P / little p distinction at 2:48 and the overview at 9:09 are worth a minute each). The lecture is about novels and films. He says a short story must hit its promise harder, in the first paragraph; for a two-minute spoken story that means the first line or two. Contains spoilers for While You Were Sleeping and the Lord of the Rings films.

CHECK YOURSELF · ANSWERS AT THE BACK

3.1 According to Sanderson, what is usually wrong when an audience finds a section boring?

A The language is too plain for the size of the promise
B The progress is on a different axis from what was promised
C There is no twist to reward the reader for waiting
D The section is too short to show any real progress being made

3.2 Why does Raiders of the Lost Ark open with a short adventure?

A To show the story in miniature and promise what the film will be
B To give the villain's backstory before the main plot begins
C To set up a twist that reverses the meaning of the film's ending
D To prove the hero always wins, so the audience can relax

3.3 What makes a payoff feel surprising yet inevitable, in the lecture?

A Hiding the promise completely until the final chapter
B Putting in a twist the audience could not possibly have seen coming
C Making them doubt they'll get what was promised, then delivering it
D Ending just before the promise is fulfilled, leaving it open

MATH 130 · PROBABILITY · LECTURE 4 OF 14

Gambler's Ruin and Random Variables

Mon 9 Nov · 52 min video · Watch on YouTube

THE TAKEAWAY

Condition on the FIRST STEP and a problem with no obvious end becomes a difference equation — and a random variable is not a variable at all but a function from the sample space to the real line.

THE ARGUMENT · 14 STEPS

1. **The setup**
   Two gamblers trade one dollar per round, A winning each round with probability p and B with $q = 1 - p$. A starts with i, B with $N - i$, and the game ends when someone is broke. Nothing is created or destroyed — N dollars move around until one player holds them all.

2. **It is a random walk with absorbing states**
   Track A's fortune on a line from 0 to N: it steps right with probability p, left with q, and sticks forever at either end. Recognising that a problem IS this problem, in different clothing, is most of the skill.

3. **First-step analysis**
   After one round the game is the same game from $i + 1$ or $i - 1$. So condition on who won the first round: $p_i = p\,p_{i+1} + q\,p_{i-1}$, with boundary conditions $p_0 = 0$ and $p_N = 1$.

4. **Solve the MORE general problem**
   Being told A starts with exactly \$100 sounds easier and is harder. The insight is the relation between p_i, p_{i+1} and p_{i-1}, which only exists if i is left general.

5. **Difference equations, solved by guessing a power**
   Try $p_i = x^i$: it forces $px^2 - x + q = 0$, whose roots are $x = 1$ and $x = q/p$ — the discriminant $1 - 4pq$ being exactly $(2p - 1)^2$. The general solution is a linear combination of the roots raised to the i.

6. **Then the boundary conditions pin it down**
   $p_i = \dfrac{1 - (q/p)^i}{1 - (q/p)^N}$ when $p \neq q$. For the fair game the two roots coincide, so take the limit by L'Hopital and get the much friendlier $p_i = i/N$.

7. **The fair answer is memorable**
   Your chance of winning everything is the fraction of the money you started with. Two thirds of the pot means a two-thirds chance.

8 **The unfair answer is brutal**

With $p = 0.49$ and both players starting equal: $N = 20$ gives 40%, $N = 100$ gives 12%, $N = 200$ gives 1.8%. A one-percent edge per round compounds into near-certainty over enough rounds.

9 **Which is why it is called gambler's ruin**

In a casino the house has far more money than you AND the games range from slightly to very unfavourable. Both of the things that hurt in this calculation are against you at once.

10 **The game ends with probability 1**

Nothing in the equations forbids oscillating forever — but $P(A \text{ wins}) + P(B \text{ wins}) = \frac{i}{N} + \frac{N-i}{N} = 1$, leaving no probability over for it. That falls out for free, with no separate argument.

11 **What a random variable actually is**

Not 'a variable taking random values' — that says nothing. It is a FUNCTION from the sample space S to the real line: a numerical summary of some aspect of the experiment. The randomness lives in the experiment, not in the function.

12 **Bernoulli**

$X \sim \text{Bern}(p)$ takes only the values 1 and 0, with probabilities p and $1 - p$. And '$X = 1$' is an EVENT — formally $\{s \in S : X(s) = 1\}$ — which is why writing $P(X = 1)$ is legitimate at all.

13 **Binomial**

$X \sim \text{Bin}(n, p)$ counts successes in n independent Bernoulli(p) trials, with $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$: choose where the successes go, then fix each trial. That function is the probability mass function.

14 **And a story proof for sums**

If $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ are independent then $X + Y \sim \text{Bin}(n + m, p)$ — because X counts successes in the first n trials and Y in the next m, so the sum counts successes in $n + m$. No algebra at all.

WORKED EXAMPLE

When a process repeats itself with a shifted starting point, condition on the first step. The recursion you get relates unknowns to unknowns, which looks circular and is not — the boundary conditions close it.

WATCH FOR

Fixing the numbers too early. Specialising i to 100 removes the only structure the problem has; generality is what makes the recursion available.

CHECK YOURSELF · ANSWERS AT THE BACK

4.1 Why does solving gambler's ruin require leaving the starting stake general?

A To make the algebra prettier
B Because the method relates p_i to $p_{i\pm1}$, and that relation does not exist for a single fixed i
C Because N is unknown
D Because the game might not end

4.2 With $p = 0.49$, both players starting at \$50, A's chance of winning everything, to three decimal places, is:

Your answer: _____

4.3 For $X \sim \text{Bin}(7, 0.5)$, $P(X = 3)$, to four decimal places, is:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 5 OF 14

Transposes, Permutations, Spaces R^n

Tue 10 Nov · 48 min video · Watch on YouTube

THE TAKEAWAY

A subspace is any set of vectors you cannot escape by adding or scaling, which is why every one of them contains the zero vector.

THE ARGUMENT · 6 STEPS

1. **Row exchanges turn $A = LU$ into $PA = LU$**
   If a zero lands in a pivot position, a permutation P puts the rows into an order that works, and elimination proceeds as before. There are $n!$ such matrices, and each one's inverse is its transpose: $P^TP = I$.

2. **Transposing swaps the indices**
   $(A^T)_{ij} = A_{ji}$ — a reflection across the main diagonal. A matrix unchanged by it is symmetric, and symmetric matrices will turn out to be the best-behaved ones in the course.

3. **R^TR is symmetric whatever R is**
   Transpose it: the order reverses, giving $R^T(R^T)^T$, and transposing twice returns R. So the product equals its own transpose. Two lines, no arithmetic, and the same two cover RR^T.

4. **A space is closed under combinations**
   Adding two members stays inside; scaling a member by any real number stays inside. The first quadrant of the plane fails: it is fine under addition, but multiplying $(3, 2)$ by -5 leaves it immediately.

5 **Which forces the zero vector in**

Any member times 0 must still be a member. So a line through the origin is a subspace of $\mathbb{R}^2$ and a parallel line beside it is not — and the whole plane, every line through the origin, and $\{\vec{0}\}$ alone are the only three kinds.

6 **A matrix gives you one for free: the column space**

Take the columns, take every linear combination of them, and what you get is closed by construction. Two independent columns in $\mathbb{R}^3$ fill a plane through the origin — not all of $\mathbb{R}^3$, and more than a line.

WORKED EXAMPLE

$R = \begin{pmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{pmatrix}$. Row 1 of RR^T against column 3 is $1 \cdot 4 + 3 \cdot 1 = 7$; row 3 against column 1 is $4 \cdot 1 + 1 \cdot 3 = 7$ as well. Same two numbers, multiplied in the other order.

WATCH FOR

Calling a line in $\mathbb{R}^2$ a copy of $\mathbb{R}^1$. Its vectors have two components, so it lives inside $\mathbb{R}^2$; only a line through the origin is even a subspace.

KEY CONCEPTS

Span is everywhere you can reach

CHECK YOURSELF · ANSWERS AT THE BACK

5.1 The first quadrant of the plane is not a subspace because:

A It does not contain the zero vector
B Multiplying a member by a negative number leaves it
C Adding two members leaves it
D It is not a plane

5.2 For $R = \begin{pmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{pmatrix}$, the entry of RR^T in row 1, column 3 is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 6 OF 14

Let's build the GPT Tokenizer

Tue 10 Nov – Thu 12 Nov · day 1–3 of 3 · 134 min video · Watch on YouTube

THE TAKEAWAY

A tokenizer is a separate model with its own training set and algorithm: byte-level BPE starts from the 256 bytes and repeatedly merges the most frequent adjacent pair, encode() replays those merges in the order they were learned, and many of an LLM's weird behaviours trace back to the chunks this produces.

THE ARGUMENT · 8 STEPS

1. **A separate stage**
   The tokenizer translates between strings and tokens (chunks of text) and is a completely separate stage of an LLM: it has its own training set and its own training algorithm, Byte Pair Encoding, and once trained it implements just encode() and decode(). A lot of an LLM's weird behaviours trace back to it.

2. **BPE, popularised by GPT-2**
   Byte Pair Encoding was popularised for LLMs by the GPT-2 paper and code release (Sennrich et al. 2015 is the original reference for BPE in NLP), and GPT, Llama and Mistral all train their tokenizers with it. minbpe implements the three functions — train, encode, decode — in a BasicTokenizer and a RegexTokenizer, plus a GPT4Tokenizer that reproduces tiktoken's GPT-4 encoding.

3. **Byte-level**
   Encode the text as UTF-8 and start from the 256 possible byte values as the first tokens, so any string in any language — emoji included — can be represented, with no unknown token.

4. **Training: merge the most frequent pair**
   Count every adjacent pair (get_stats), mint a new id for the most common (256, 257, …), replace every occurrence (merge), repeat vocab_size − 256 times. On 'aaabdaaabac' minbpe's three merges are aa → 256, then (256, a) → 257 ('aaa'), then (257, b) → 258 ('aaab'): after the first merge two pairs tie at two each, and max() keeps the one counted first. The string becomes [258, 100, 258, 97, 99]. The README's walk-through (aa, then ab, then aa+ab) is Wikipedia's tie-break, not the code's; the ids come out the same.

5. **Encode and decode**
   decode() concatenates each id's bytes and decodes as UTF-8 with errors='replace', because a token sequence can be invalid UTF-8. encode() starts from bytes and repeatedly applies the available pair with the LOWEST merge index — the order the merges were learned — until none applies.

6. **Regex splitting, as GPT-2 and GPT-4 do**
   RegexTokenizer first splits the text with a pattern into letters, numbers (GPT-4: at most three digits), punctuation and whitespace, and runs BPE inside each chunk, so no token crosses a category boundary. Trained on enough data at 100K vocab, it should reproduce GPT-4's tokenizer.

7. **Special tokens**
   Tokens such as end-of-text (id 100257 in GPT-4) are registered after the merges and inserted by splitting on exact matches. encode() refuses them by default — allowed_special must be set — because tokenizing attacker-controlled text as control tokens is a footgun.

8. **Matching GPT-4 exactly**
   GPT4Tokenizer loads tiktoken's cl100k_base ranks, recovers the merges from them, and undoes a historical permutation of the 256 byte tokens; its tests check identical ids to tiktoken. sentencepiece, used by Llama and Mistral, runs BPE on Unicode code points rather than bytes.

WORKED EXAMPLE

To see what a model sees, run the text through its tokenizer and print the pieces with their ids. Look for leading spaces, numbers split at odd places, and words in other languages broken into many byte-level pieces: whatever looks arbitrary there is what the network has to learn around.

WATCH FOR

Tokenizing untrusted text with special tokens enabled. If a user's message contains the literal end-of-text string and it becomes the real control token, the user can end documents or impersonate structure — which is why encode() refuses unless allowed_special is set deliberately.

CHECK YOURSELF · ANSWERS AT THE BACK

6.1 BasicTokenizer is trained on 'aaabdaaabac' with vocab_size = 256 + 3, then encodes the same string. How many tokens does it produce?

Your answer: _____

6.2 A RegexTokenizer is trained with vocab_size = 32768. How many merges does it learn? (digits only, no commas)

Your answer: _____

6.3 Why does a byte-level BPE tokenizer never meet a character it cannot encode?

A Every string is UTF-8 bytes, and all 256 bytes are already tokens
B Its training text is large enough to contain every possible character
C It maps every unknown character to one shared special unknown token
D It falls back to splitting the text into individual Unicode code points

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 7 OF 14

Chain Rule

Wed 11 Nov · 46 min video · Watch on YouTube

THE TAKEAWAY

Product, quotient and chain rules are proved the same way — write the change, add and subtract a middle term, divide by Δx — and the chain rule is the one that sets you free.

THE ARGUMENT · 6 STEPS

1 **The product rule comes from changing one factor at a time**
Write $\Delta(uv)$, then add and subtract $u(x)v(x+\Delta x)$. The middle terms cancel and what is left is $\Delta u \cdot v(x+\Delta x) + u(x) \cdot \Delta v$. Divide by Δx and let it go: $(uv)' = u'v + uv'$.

2 **Continuity does the last step**
In the limit, $v(x+\Delta x) \to v(x)$ — and that is exactly the theorem that differentiable implies continuous. It is where that earlier result gets used.

3 **The quotient rule, by the same method**
Put the difference over a common denominator, watch uv cancel twice, and $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$. Ugly, and it has to be memorised in the right order, because the numerator is not symmetric.

4 **Which quietly extends the power rule**
Take $u = 1$ and $v = x^n$: the quotient rule gives $-nx^{-n-1}$. That is the SAME formula $\frac{d}{dx}x^m = mx^{m-1}$, now with m negative. One rule, one proof, twice the range.

5 **The chain rule is almost algebra**
$\frac{\Delta y}{\Delta t} = \frac{\Delta y}{\Delta x} \cdot \frac{\Delta x}{\Delta t}$ — the middle change genuinely cancels. In the limit, $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$. Naming the inner value x makes it obvious; with practice you skip the name.

6 **Higher derivatives, and $D^n x^n = n!$**
Differentiating x^n repeatedly brings the exponent down each time: n, then $n-1$, down to 1. After n passes the product of those is $n!$ and no x remains — so the next derivative is 0.

WORKED EXAMPLE

$\frac{d}{dt}(\sin t)^{10}$. Outside is the tenth power, inside is $\sin t$. Differentiate the outside at the inside value: $10(\sin t)^9$. Multiply by the inside's derivative: $\cos t$. Answer $10\sin^9 t\cos t$.

WATCH FOR

Reversing the quotient rule's numerator. It is $u'v - uv'$, and $uv' - u'v$ is its negative — a sign error that survives every later step. The product rule is symmetric; this one is not.

CHECK YOURSELF · ANSWERS AT THE BACK

7.1 The power rule holds for negative exponents because:

A It was assumed from the start
B Applying the quotient rule to $1/x^n$ produces exactly mx^{m-1} with $m = -n$
C Negative powers are undefined
D The chain rule covers it

7.2 The fifth derivative of x^5 is the constant:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 8 OF 14

Homework for Life, sharpened — and the pickle story

Wed 11 Nov · 7 min video · Watch on YouTube

THE TAKEAWAY

Dicks revisits Homework for Life: the daily note gets sharper with practice, and the bank pays off when you need a story for a message — you match a small personal moment to it by adjacency instead of borrowing someone else's.

THE ARGUMENT · 6 STEPS

1. **The person with the most stories wins**
   The best storytellers have the most stories to choose from. He started Homework for Life because he was running out.

2. **The kidnapping prompt**
   If his family would only be returned for a story about today, which moment would he tell? Two columns: the date, and a line or two.

3. **The lens sharpens**
   In 2015 he noticed 1.3 moments a day, mostly things he did. Last year it was 7.8 — including things heard, said or only thought — because the lens got sharper, not his life.

4. **Why time flies**
   Most people remember perhaps fifty days of a year with any clarity. A daily record turns a year back into 365 days.

5. **Matching by adjacency**
   A contract manager wanted his team to be more specific. Dicks hates pickles and always tells servers not to put one on the plate — a story about being specific.

6. **Don't borrow**
   The manager wanted to use the pickle story himself. It would land the message but reveal nothing about him, and revealing yourself is what makes people connect.

WORKED EXAMPLE

Write the message you need ('be specific', 'ask before you build'). Scan your bank for a small personal moment where you behaved that way about something unrelated. Tell the moment, then bridge: 'that is how I want us to be about X.'

WATCH FOR

Borrowing a good story from someone else. It may land, but it costs you the thing a story is for: the audience learning something true about the person in front of them.

CHECK YOURSELF · ANSWERS AT THE BACK

8.1 How many moments a day was Dicks noticing in 2015, compared with recently?

A About 1.3 then, about 7.8 now
B About 5 then, about 5 now
C About 10 then, about 3 now
D One a week then, one a day now

8.2 What does matching a story 'by adjacency' mean?

A Telling your stories in the order they happened, one after another
B Choosing a story from the same industry as the message you need
C Using a small personal moment that mirrors the message's behaviour
D Placing two stories side by side so the audience compares them

8.3 Why shouldn't the manager simply use Dicks's pickle story?

A It belongs to Dicks, and a borrowed story always sounds second-hand
B It is about restaurants, so his team wouldn't see the connection to contracts
C It reveals nothing about the manager, which is where a story's power is
D It is too long to tell in a team meeting about contracts

SPCH 100 · STORYTELLING · LECTURE 9 OF 14

Seeing with storyteller eyes — notice the flag

Wed 11 Nov · 3 min video · Watch on YouTube

THE TAKEAWAY

No life is too boring for stories; the difference is noticing. The observant teller treats the mind's small flags — frustration, surprise, laughter — as stories arriving.

THE ARGUMENT · 5 STEPS

1. **Boring life syndrome**
   The comforting belief that great speakers simply have more interesting lives. It is an excuse.

2. **No boring stories, only boring storytellers**
   A rough truth, and an important one.

3. **The friend with the ordinary life**
   Work, a baby, cafés, the daycare, cycling home — and always a story about the coffee queue or the barista, because she was observant.

4. **Believe you are a storyteller**
   Deciding that you are one is what changes what you see.

5 The flag

When you are frustrated, surprised or laughing, your mind raises a flag that says pay attention. Most of the time we let those moments fly past.

WORKED EXAMPLE

At the end of a day, list the moments your mind raised a flag — irritation, surprise, a laugh — however trivial. Those, not the day's events, are your raw material.

WATCH FOR

Looking for events. The observant friend's days held no events at all; what she had was a habit of noticing.

CHECK YOURSELF · ANSWERS AT THE BACK

9.1 What is 'boring life syndrome'?

A A fear that your own life is too dull to interest an audience
B The belief that good speakers just have more interesting lives
C Telling the same few stories so often that they go stale
D Being so bored at work that nothing seems worth noticing

9.2 Which of these is the 'flag' the speaker describes?

A A sudden moment of frustration, surprise or laughter
B A notification that someone has mentioned you online
C A deadline that forces you to look back over your week
D A compliment from your boss that you want to remember

9.3 Why did the friend with an ordinary life have so many stories?

A She travelled more than most people and met strangers
B She worked as a writer and practised every single day
C She was observant and saw the world as a storyteller
D She exaggerated small events until they sounded big

SPCH 100 · STORYTELLING · LECTURE 10 OF 14

A photo folder called Stories

Wed 11 Nov · 5 min video · Watch on YouTube

THE TAKEAWAY

A speaker's simplest capture habit: when something strikes you as curious, odd or frustrating, take a photo into a folder called Stories. When a talk needs an illustration, scan the folder.

THE ARGUMENT · 5 STEPS

1. **Three photos, no point**
   A hotel suggestion box with no paper or pens, a rental car tyre botched with glue and four and a half hours stranded, a cooking class on mole with up to a hundred ingredients.

2. **The point is the habit**
   Each was a moment that sparked curiosity or frustration, and each took two seconds to capture.

3. **The Stories folder**
   When preparing a talk or a video and the point needs an illustration, scan the folder of hundreds of moments.

4. **Walking story banks**
   We all have thousands of stories; we just forget to record them. 'I don't have a story' is not believable.

5. **Business runs on stories**
   Presentations, interviews and pitches all tell a story about a product, a person or an idea.

WORKED EXAMPLE

Make the capture smaller than the decision: photograph first, decide later. When preparing anything, open the folder before you open a blank page.

WATCH FOR

Trying to remember the moment instead of recording it. The flat tyre feels unforgettable for a week and is gone in a month.

CHECK YOURSELF · ANSWERS AT THE BACK

10.1 What should go into the Stories folder?

A Only photos of events important enough to build a talk on
B Anything that piques your curiosity, interest or frustration
C Photos of audiences, so you remember who laughed and when
D Slides from other speakers' talks that you thought worked

10.2 When is the folder used?

A When preparing a talk or video and a point needs a story
B At the end of each year, to choose the best story to tell
C Every week, to post the best photo straight to social media
D Never; capturing the moment is the whole of the exercise

10.3 Why does the speaker not believe clients who say they have no stories?

A Everyone has been on holiday, and holidays are full of stories
B Clients exaggerate how little has happened to them, out of modesty
C We all have thousands of stories and simply fail to record them
D A good speaker can invent a story when no true one is available

MATH 130 · PROBABILITY · LECTURE 11 OF 14

Random Variables and Their Distributions

Thu 12 Nov · 50 min video · Watch on YouTube

THE TAKEAWAY

A random variable and its DISTRIBUTION are different objects — the house and the blueprint — and the distribution is carried either by the PMF, which says what each value weighs, or by the CDF, which says how much weight is at or below each point.

THE ARGUMENT · 13 STEPS

1 **Three ways to see the binomial, and you need all three**
The STORY — successes in n independent Bernoulli(p) trials. The SUM — $X = X_1 + \cdots + X_n$ of IID indicators. The PMF — $P(X = k) = \binom{n}{k} p^k q^{n-k}$. Each one makes some questions trivial and others impossible, so the skill is switching between them.

2 **IID**
Independent and identically distributed. Two separate claims: identically distributed says the blueprints match, independent says knowing one tells you nothing about another. Either can hold without the other.

3 **The distribution is not the random variable**
Many different random variables can share one distribution, the way many houses can be built from one blueprint. X and Y can both be Bin(n, p) and still never be equal. Confusing the two is the single most common mistake in the course.

4 **The CDF**
$F(x) = P(X \le x)$, defined for every real x even when X only takes integer values. It works for any random variable, discrete or not, which is why the theory does not have to be split in two.

5 **What the PMF has to satisfy**
Exactly two conditions: $p_j \ge 0$, and $\sum_j p_j = 1$. Nothing else. Any list of numbers meeting those two is somebody's PMF.

6 **And the binomial does satisfy them**
$\sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k} = (p+q)^n = 1$ by the binomial theorem. The validity check is the binomial theorem itself, which is a good sign that the PMF was not invented arbitrarily.

7 **Bin(n, p) + Bin(m, p) = Bin($n + m, p$), proved three ways**
By STORY: n trials then m more trials is $n + m$ trials, done in one line. By INDICATORS: a sum of n Bernoullis plus a sum of m Bernoullis is a sum of $n + m$ Bernoullis. By ALGEBRA: condition on X and sum, which reduces to Vandermonde's identity. The story proof is the one to keep.

8 **Five cards is NOT binomial**

Count the aces in a five-card hand and the trials are not independent — if the first four cards are aces the fifth cannot be. Sampling WITHOUT replacement breaks the binomial's only real assumption.

9 **The hypergeometric**

$P(X = k) = \frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}}$: choose which of the tagged items you got, which of the untagged, over all ways to choose n from the pooled population.

10 **One distribution, three stories**

Tagged and untagged elk recaptured; white and black marbles drawn without replacement; aces and non-aces in a hand. Recognising that these are the SAME problem is most of what naming a distribution buys you.

11 **Vandermonde again**

The hypergeometric PMF sums to 1 because $\sum_k \binom{w}{k}\binom{b}{n-k} = \binom{w+b}{n}$ — the same identity that appeared in the algebraic proof for the binomial, now doing the validity check.

12 **When the population is huge, it is binomial anyway**

Drawing 5 cards from 52 changes the composition noticeably; drawing 5 items from ten million does not. With replacement and without replacement converge, which is why the binomial gets used for sampling problems it does not strictly fit.

13 **What the two CDF pictures look like**

Continuous: a smooth curve, increasing, running from 0 up to 1. Discrete: a staircase with a jump at each possible value, closed circle on top and open circle below, and the height of each jump IS the PMF at that value.

WORKED EXAMPLE

Before computing anything, decide which of the three views of the distribution the question is asking about — the story, the sum, or the formula. Most of the difficulty in a distribution problem is having picked the wrong one of the three.

WATCH FOR

Treating a count drawn without replacement as binomial. The formula will produce a number; it will be the wrong number, and nothing in the arithmetic will complain.

CHECK YOURSELF · ANSWERS AT THE BACK

11.1 Why is the number of aces in a five-card hand not binomial?

A Because there are only four aces
B Because the trials are dependent — what the earlier cards were changes the odds for the later ones
C Because five is too small a sample
D Because the probability of an ace is not $1/13$

11.2 The probability that a five-card hand contains exactly two aces, to four decimal places, is:

Your answer: _____

11.3 If you wrongly modelled that same count as Bin$(5, 1/13)$, the probability of exactly two aces, to four decimal places, would be:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 12 OF 14

First, last, best, worst — mining the past

Thu 12 Nov · 7 min video · Watch on YouTube

THE TAKEAWAY

Two exercises for an empty story bank: Homework for Life to catch new moments as they happen, and First-Last-Best-Worst to dig old ones out of the past.

THE ARGUMENT · 5 STEPS

1. **Homework for Life, briefly**
   Every day ask which moment you would tell a story about; write the date, the moment, and optionally a lesson. Early entries feel trivial — that is normal.

2. **What it gives you**
   A pool of stories, more mindfulness as you walk around, and patterns: you start to see what triggers you and how you react.

3. **The grid**
   Across the top: first, last, best, worst. Down the side: prompts such as gift, car, teacher, pet, holiday.

4. **One or two words per cell**
   Do not write stories yet; just get the memories down.

5 **Then choose**

Scan the table for the memories that make you think 'that could be a story' — roughly half will be new ideas.

WORKED EXAMPLE

Fill the grid fast, without judging, in one sitting. Then star two cells and run each through the five-second-moment test: what changed in me?

WATCH FOR

Writing full stories into the grid. It slows you down, you fill four cells instead of twenty, and the best memories surface late.

CHECK YOURSELF · ANSWERS AT THE BACK

12.1 What are the four column headings of the grid?

A Who, what, where, when
B First, last, best, worst
C Past, present, future, never
D Happy, sad, angry, afraid

12.2 How much should you write in each cell?

A A full paragraph, so you don't lose the details
B Only the date, and fill the rest in later
C One or two words to remember the memory
D Nothing at all until you can name the lesson

12.3 How do the two exercises differ?

A Homework for Life catches new moments; the grid digs out past ones
B Homework for Life is for business stories; the grid is for personal ones
C The grid is done once a week; Homework for Life is done every month
D The grid catches new moments; Homework for Life digs into the past

MATH 110 · LINEAR ALGEBRA · LECTURE 13 OF 14

Column Space and Nullspace

Fri 13 Nov · 46 min video · Watch on YouTube

THE TAKEAWAY

The column space says which right-hand sides are reachable; the null space says how many ways each reachable one can be reached.

THE ARGUMENT · 6 STEPS

1. **A subspace is closed, and closure is the whole test**
   Add two members, scale a member by any real number, and stay inside. A plane through the origin in $\mathbb{R}^3$ passes; the union of a plane and a line through it fails, because one vector from each adds to something in neither.

2. **An intersection is always a subspace**
   If $\vec{v}$ and $\vec{w}$ lie in both S and T, their sum lies in S and in T, so it lies in both. The same argument handles scaling. Intersections shrink the set and keep the structure; unions do the reverse.

3. **The column space is every combination of the columns**
   For a four-by-three matrix the columns live in $\mathbb{R}^4$, so $C(A)$ is a subspace of $\mathbb{R}^4$ — but three vectors cannot fill four dimensions, so it is a proper subspace and most of $\mathbb{R}^4$ is out of reach.

4. **$A\vec{x} = \vec{b}$ is solvable exactly when $\vec{b}$ is in $C(A)$**
   That is not a theorem so much as a restatement: $A\vec{x}$ IS a combination of the columns. If $\vec{b}$ is one, the recipe is $\vec{x}$; if it is not, no $\vec{x}$ exists. Four equations in three unknowns are unsolvable for most $\vec{b}$ and solvable for some.

5. **A dependent column adds nothing**
   In $\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{pmatrix}$ the third column is the first plus the second. Throw it away and the column space is unchanged: a plane in $\mathbb{R}^4$, not a three-dimensional space.

6. **The null space collects the $\vec{x}$ with $A\vec{x} = \vec{0}$**
   Here it is every multiple of $(1, 1, -1)$ — a line in $\mathbb{R}^3$. It is a subspace because $A\vec{v} = A\vec{w} = \vec{0}$ forces $A(\vec{v} + \vec{w}) = \vec{0}$. The solutions of $A\vec{x} = \vec{b}$ for non-zero $\vec{b}$ are not, since $\vec{0}$ is not among them.

WORKED EXAMPLE

Two ways to name a subspace, and the two spaces use one each. Hand over some vectors and say take all combinations — that builds the column space. Hand over equations and say find everything that satisfies them — that carves out the null space.

WATCH FOR

Testing for a subspace by checking addition only. The first quadrant of the plane is closed under addition and still fails, because scaling by a negative number leaves it.

KEY CONCEPTS

Span is everywhere you can reach
Rank plus nullity is conserved

CHECK YOURSELF · ANSWERS AT THE BACK

13.1 The union of a plane and a line, both through the origin in $\mathbb{R}^3$, is not a subspace because:

A It omits the zero vector
B A vector from the plane plus one from the line usually lies in neither
C It is not closed under scaling
D A plane and a line cannot intersect

13.2 For $A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{pmatrix}$, the dimension of the column space is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 14 OF 14

Let's reproduce GPT-2 (124M)

Fri 13 Nov · day 1 of 5 · 241 min video · Watch on YouTube · finishes in week 5

THE TAKEAWAY

Reproducing GPT-2 (124M) is the GPT from before plus everything that makes it real: the exact architecture so OpenAI's weights load, GPT-2's initialisation, a long list of speed-ups, GPT-3's optimisation recipe — AdamW, clipping, warmup and cosine decay, a 0.5M-token batch via gradient accumulation — and honest evaluation on held-out loss and HellaSwag.

THE ARGUMENT · 8 STEPS

1. **Match GPT-2 exactly, then load it**
   12 layers, 12 heads, n_embd 768, context 1,024, vocabulary 50,257 (50,000 merges + 256 bytes + end-of-text). Module names follow Hugging Face's, so from_pretrained copies OpenAI's weights in — transposing the Conv1D ones — and the model samples sensible text before any training.

2. **Data as shifted token buffers**
   A buffer of B·T + 1 tokens becomes x (all but the last) and y (all but the first), viewed as (B, T). The loss is cross-entropy over every position. A tiny batch can be overfitted to check the pipeline before training for real.

3. **Weight tying and GPT-2's init**
   The token embedding wte and the output lm_head share one matrix — GPT-2's checkpoint already does. Weights start at std 0.02, and the projections that write into the residual stream are scaled by $1/\sqrt{2 \cdot n_{\text{layer}}}$, because each block adds two contributions and the stream's std would otherwise grow with depth.

4 **Make it fast**

TF32 matmuls, bfloat16 autocast, torch.compile, flash attention through F.scaled_dot_product_attention, and the vocabulary padded from 50,257 to the 'nice' 50,304 — each its own commit, each measured in tokens per second.

5 **GPT-3's optimisation recipe**

AdamW with betas (0.9, 0.95) and eps 1e-8, gradient norm clipped at 1.0, a learning rate that warms up linearly for 715 steps to 6e-4 and then decays on a cosine to 6e-5, and weight decay 0.1 on 2-D parameters only.

6 **A 0.5M-token batch on any hardware**

total_batch_size is 524,288 tokens; with micro-batches of B·T tokens the gradient is accumulated over several forward/backward passes before one optimizer step, and the loss is divided by the number of micro-steps so the accumulated gradient is a mean, not a sum. DDP splits the micro-steps across GPUs and averages.

7 **Real data: FineWeb-Edu**

The 10B-token sample, tokenized with GPT-2's tokenizer, end-of-text between documents, stored as uint16 in 100 shards of 100M tokens, the first shard kept for validation. 19,073 steps of 0.5M tokens is about one epoch.

8 **Evaluate honestly**

Validation loss every 250 steps, and HellaSwag in completion style: each of the four endings is appended to the context and the one with the lowest average per-token loss is chosen. The 124M run is compared with OpenAI's GPT-2 checkpoint (val 3.2924; HellaSwag 0.2945) and GPT-3 124M (0.337).

WORKED EXAMPLE

When changing the batch size, keep the objective the same: fix total_batch_size in tokens, derive grad_accum_steps = total / (B · T · world_size), and divide each micro-step's loss by grad_accum_steps. Then the gradient matches the big batch's mean whatever micro-batch fits in memory — the play notebook checks exactly this on a tiny MLP.

WATCH FOR

Accumulating gradients without rescaling the loss. backward() adds gradients, so k micro-steps produce the gradient of a SUM of k mean losses — k times too large — which silently multiplies the effective learning rate by k.

CHECK YOURSELF · ANSWERS AT THE BACK

14.1 GPT-2 (124M) ties its token embedding to its output layer: vocab 50,257, n_embd 768. How many parameters does sharing that one matrix save? (digits only, no commas)

Your answer: _____

14.2 _init_weights multiplies the std of every c_proj layer by (2 · n_layer)^−0.5. Why?

A To make those projection layers learn faster than the rest of the network
B Each block adds two outputs to the residual stream; this stops its std growing
C To compensate for the GELU nonlinearity, which halves the variance of its input
D Because the projection layers have no bias, so they need less noise

14.3 total_batch_size = 524,288 tokens, micro-batch B = 64 sequences of T = 1,024 tokens, on one GPU. How many gradient-accumulation micro-steps make one optimizer step?

Your answer: _____

WEEK 4 · SUN 8 – FRI 13 NOVEMBER 2026

What this week established

Each lecture’s takeaway, by course.

MATH 120 · Single-Variable Calculus

1 **Derivatives.** Both trig derivatives come from two limits at zero, and those two limits come from one geometric fact: short pieces of curve are nearly straight.

7 **Chain Rule.** Product, quotient and chain rules are proved the same way — write the change, add and subtract a middle term, divide by Δx — and the chain rule is the one that sets you free.

AI 200 · Neural Networks: Zero to Hero

2 **Let's build GPT — from scratch, in code.** A GPT is a bigram model that learned to look back: tokens gather information from earlier positions through masked, scaled self-attention, compute on it with a feed-forward net, and stack those blocks with residual connections and layer norms — each addition visibly lowering the validation loss on Tiny Shakespeare.

6 **Let's build the GPT Tokenizer.** A tokenizer is a separate model with its own training set and algorithm: byte-level BPE starts from the 256 bytes and repeatedly merges the most frequent adjacent pair, encode() replays those merges in the order they were learned, and many of an LLM's weird behaviours trace back to the chunks this produces.

14 **Let's reproduce GPT-2 (124M).** Reproducing GPT-2 (124M) is the GPT from before plus everything that makes it real: the exact architecture so OpenAI's weights load, GPT-2's initialisation, a long list of speed-ups, GPT-3's optimisation recipe — AdamW, clipping, warmup and cosine decay, a 0.5M-token batch via gradient accumulation — and honest evaluation on held-out loss and HellaSwag.

SPCH 100 · Storytelling

3 **Promise, progress, payoff — why a middle drags and an ending lands (start at 19:28).** Brandon Sanderson reduces plot to promise, progress and payoff. The opening promises what kind of story it is; the middle must keep showing movement toward that promise; the ending fulfils it in a way that feels surprising yet inevitable. When an audience is bored, the progress is usually on a different axis from the promise.

8 **Homework for Life, sharpened — and the pickle story.** Dicks revisits Homework for Life: the daily note gets sharper with practice, and the bank pays off when you need a story for a message — you match a small personal moment to it by adjacency instead of borrowing someone else's.

9 **Seeing with storyteller eyes — notice the flag.** No life is too boring for stories; the difference is noticing. The observant teller treats the mind's small flags — frustration, surprise, laughter — as stories arriving.

10 **A photo folder called Stories.** A speaker's simplest capture habit: when something strikes you as curious, odd or frustrating, take a photo into a folder called Stories. When a talk needs an illustration, scan the folder.

12 **First, last, best, worst — mining the past.** Two exercises for an empty story bank: Homework for Life to catch new moments as they happen, and First-Last-Best-Worst to dig old ones out of the past.

MATH 130 · Probability

4 **Gambler's Ruin and Random Variables.** Condition on the FIRST STEP and a problem with no obvious end becomes a difference equation — and a random variable is not a variable at all but a function from the sample space to the real line.

11 **Random Variables and Their Distributions.** A random variable and its DISTRIBUTION are different objects — the house and the blueprint — and the distribution is carried either by the PMF, which says what each value weighs, or by the CDF, which says how much weight is at or below each point.

MATH 110 · Linear Algebra

5 **Transposes, Permutations, Spaces R^n.** A subspace is any set of vectors you cannot escape by adding or scaling, which is why every one of them contains the zero vector.

13 **Column Space and Nullspace.** The column space says which right-hand sides are reachable; the null space says how many ways each reachable one can be reached.

WEEK 4 · SUN 8 – FRI 13 NOVEMBER 2026

Answers

The answers and explanations are the summaries' own.

1 Derivatives

1.1 B — The proof compares the arc length θ with a chord, so the angle must be measured as length on the unit circle

The limit $\frac{\sin\theta}{\theta} \to 1$ says a chord and an arc have the same length in the limit. That is a statement about two lengths, and it fails if θ is counted in 360ths of a turn.

1.2 0.5

It is $\cos(\pi/3) = \frac{1}{2}$. The sine graph is still rising there, but less steeply than at 0, where the derivative is 1.

2 Let's build GPT — from scratch, in code

2.1 8

Position t of x, with everything before it, predicts y[t]: contexts of length 1, 2, … up to 8, each with its own target. Training on every length is also what lets the model generate from a context as short as one token.

2.2 C — So each position gathers only from itself and earlier positions, never later

exp(−inf) = 0, so after the softmax every future position gets exactly zero weight and each row still sums to 1 over the allowed ones. A token predicting the next character must not see it — this is a decoder; an encoder would simply leave the mask out.

2.3 0.125

head_size = n_embd // n_head = 384 / 6 = 64, and $64^{-1/2} = 0.125$. Scaling by the head size keeps the scores' variance near 1, so the softmax stays diffuse at initialisation instead of turning nearly one-hot. The gpt.py published with the video multiplied by C**-0.5, where C is the embedding size — $384^{-1/2} \approx 0.051$; a community fix merged on 27 January 2023, ten days after the video, changed it to the head size.

3 Promise, progress, payoff — why a middle drags and an ending lands (start at 19:28)

3.1 B — The progress is on a different axis from what was promised

His usual fix is to change the promise so it matches the progress.

3.2 A — To show the story in miniature and promise what the film will be

The rest of the film is the same shape at length.

3.3 C — Making them doubt they'll get what was promised, then delivering it

Obstacles, escalation and red herrings create the doubt.

4 Gambler's Ruin and Random Variables

4.1 B — Because the method relates p_i to $p_{i\pm1}$, and that relation does not exist for a single fixed i

First-step analysis produces a recursion, and a recursion needs a family of unknowns. Pinning i down leaves one number with nothing to be related to.

4.2 0.119

$\frac{1-(q/p)^{50}}{1-(q/p)^{100}} = 0.1192$. A one-point disadvantage per round becomes an 88% chance of ruin.

4.3 0.2734

$\binom{7}{3}(0.5)^3(0.5)^4 = 35/128 = 0.2734$ — choose which three of the seven trials succeeded, then fix every trial.

5 Transposes, Permutations, Spaces R^n

5.1 B — Multiplying a member by a negative number leaves it

It contains $\vec{0}$ and it is closed under addition. What it fails is scaling: $-5 \cdot (3,2) = (-15,-10)$ is in the opposite quadrant, so the set is not closed under multiplication by every real number.

5.2 7

$(1,3)\cdot(4,1) = 4+3 = 7$. The 3,1 entry is $(4,1)\cdot(1,3)$, the same dot product — which is the whole reason RR^T is symmetric.

6 Let's build the GPT Tokenizer

6.1 5

train() takes max(stats, key=stats.get), and on a tie max() keeps the pair counted first. Merge 1: aa → 256 (four occurrences). Now (256, a) and (a, b) both occur twice, and (256, a) was counted first, so merge 2: aaa → 257. Merge 3: (257, b), i.e. aaab → 258. The string becomes 258, d, 258, a, c — [258, 100, 258, 97, 99], five tokens for eleven bytes. The README and the test's docstring explain it with Wikipedia's merges (aa, ab, then aa+ab, 'XdXac'); the code reaches the same ids by a different second merge.

6.2 32512

The first 256 ids are the raw bytes, which need no training, so num_merges = vocab_size − 256 = 32,512. The last merge gets id 32,767, and the README notes a first special token should therefore take id 32,768.

6.3 A — Every string is UTF-8 bytes, and all 256 bytes are already tokens

The vocabulary always starts with bytes 0–255, and any text — Korean, emoji, anything — is a sequence of UTF-8 bytes. Merges only shorten the sequence; the bytes are always there to fall back on. Splitting into code points is what sentencepiece does instead.

7 Chain Rule

7.1 B — Applying the quotient rule to $1/x^n$ produces exactly mx^{m-1} with $m = -n$

With $u = 1$, the quotient rule gives $-v'/v^2 = -nx^{n-1}/x^{2n} = -nx^{-n-1}$ — the same formula with the exponent negative in both places. Nothing new had to be assumed.

7.2 120

Each pass drops the exponent in front: $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5! = 120$. The sixth derivative is 0, since a constant differentiates to nothing.

8 Homework for Life, sharpened — and the pickle story

8.1 A — About 1.3 then, about 7.8 now

His life did not get more interesting; the lens got sharper, and he now counts things heard, said or thought.

8.2 C — Using a small personal moment that mirrors the message's behaviour

The pickle story is about restaurants, not contracts, but it carries exactly the behaviour he wants.

8.3 C — It reveals nothing about the manager, which is where a story's power is

The more we reveal about ourselves, the more connected the audience gets.

9 Seeing with storyteller eyes — notice the flag

9.1 B — The belief that good speakers just have more interesting lives

It is seductive because it excuses us from noticing our own material.

9.2 A — A sudden moment of frustration, surprise or laughter

Those reactions are the mind saying: pay attention, something happened here.

9.3 C — She was observant and saw the world as a storyteller

Her life was not extraordinary; her noticing was.

10 A photo folder called Stories

10.1 B — Anything that piques your curiosity, interest or frustration

The test is that it caught your attention, not that it was important.

10.2 A — When preparing a talk or video and a point needs a story

It replaces searching a blank memory under pressure.

10.3 C — We all have thousands of stories and simply fail to record them

We are walking story banks that forget to keep a record.

11 Random Variables and Their Distributions

11.1 B — Because the trials are dependent — what the earlier cards were changes the odds for the later ones

Binomial needs independent trials with a fixed success probability. Dealing without replacement gives neither, which is exactly what the hypergeometric is for.

11.2 0.0399

$\binom{4}{2}\binom{48}{3}/\binom{52}{5} = 103{,}776/2{,}598{,}960 = 0.0399$ — the hypergeometric with $w = 4$, $b = 48$, $n = 5$.

11.3 0.0465

$\binom{5}{2}(1/13)^2(12/13)^3 = 0.0465$. Seventeen per cent too high, because pretending the cards are replaced makes a second ace easier than it is.

12 First, last, best, worst — mining the past

12.1 B — First, last, best, worst

Each prompt — gift, car, teacher — is asked four ways.

12.2 C — One or two words to remember the memory

Speed matters; developing the story comes later.

12.3 A — Homework for Life catches new moments; the grid digs out past ones

One trains your eye forward, the other mines your memory.

13 Column Space and Nullspace

13.1 B — A vector from the plane plus one from the line usually lies in neither

Both pieces contain $\vec{0}$ and both are closed under scaling. Addition is what breaks: the sum steps off the plane and off the line at once.

13.2 2

Column three is column one plus column two, so it contributes nothing. The two that remain are independent, and $C(A)$ is a plane inside $\mathbb{R}^4$.

14 Let's reproduce GPT-2 (124M)

14.1 38597376

Each of the two matrices is 50,257 × 768 = 38,597,376 parameters — about 30% of the 124M model. The notebook shows OpenAI's checkpoint stores lm_head and wte with the same values at the same memory address; in the code it is one line, wte.weight = lm_head.weight.

14.2 B — Each block adds two outputs to the residual stream; this stops its std growing

Every block adds attention's and the MLP's projections into the stream, 2 · n_layer additions in all, and adding n random contributions grows the std like $\sqrt{n}$. The notebook adds 100 random vectors scaled by $100^{-1/2}$ and the std stays about 1. Hence the c_proj layers, the ones writing into the stream, are scaled down.

14.3 8

One micro-step processes 64 × 1,024 = 65,536 tokens; 524,288 / 65,536 = 8. With DDP on 8 GPUs the divisor gains a factor of 8 and it becomes 1 micro-step per GPU. The total is 2^19, about 0.5M tokens — GPT-3's batch size for this model size.